

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Symmetric Logarithmic Exponent

Evaluate the definite integral:

$$\int_{1/\sqrt{3}}^{\sqrt{3}} \frac{dx}{(1+x^2) \left(1 + (\sqrt{3}+1)^{\ln^3(\sec(\ln(x)) + \tan(\ln(x)))}\right)}$$

Solution: Let the given integral be I . We first analyze the complicated exponent in the denominator. Let $g(x) = \ln^3(\sec(\ln(x)) + \tan(\ln(x)))$. Consider the transformation $x \rightarrow \frac{1}{x}$. We evaluate $g(1/x)$:

$$\ln(1/x) = -\ln(x)$$

Substituting this into the trigonometric expression:

$$\sec(-\ln(x)) + \tan(-\ln(x)) = \sec(\ln(x)) - \tan(\ln(x))$$

Using the identity $\sec^2(\theta) - \tan^2(\theta) = 1$, we can rewrite this as:

$$\sec(\ln(x)) - \tan(\ln(x)) = \frac{1}{\sec(\ln(x)) + \tan(\ln(x))} = (\sec(\ln(x)) + \tan(\ln(x)))^{-1}$$

Thus, taking the natural logarithm:

$$\ln(\sec(-\ln(x)) + \tan(-\ln(x))) = -\ln(\sec(\ln(x)) + \tan(\ln(x)))$$

Cubing both sides preserves the negative sign, meaning $g(1/x) = -g(x)$. The function is odd with respect to the transformation $x \rightarrow 1/x$.

Let $b = \sqrt{3} + 1$. The integral is of the form $I = \int_{1/a}^a \frac{dx}{(1+x^2)(1+b^{g(x)})}$. Apply the substitution $x = \frac{1}{u}$, where $dx = -\frac{1}{u^2} du$:

$$I = \int_{\sqrt{3}}^{1/\sqrt{3}} \frac{-1/u^2}{(1+1/u^2)(1+b^{g(1/u)})} du = \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{du}{(u^2+1)(1+b^{-g(u)})}$$

Multiply the numerator and denominator by $b^{g(u)}$:

$$I = \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{b^{g(u)}}{(u^2+1)(b^{g(u)}+1)} du$$

Since u is a dummy variable, replace it with x and add this equation to the original integral I :

$$2I = \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{1}{(1+x^2)(1+b^{g(x)})} dx + \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{b^{g(x)}}{(1+x^2)(1+b^{g(x)})} dx$$

$$2I = \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{1 + b^{g(x)}}{(1 + x^2)(1 + b^{g(x)})} dx = \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{1}{1 + x^2} dx$$

This is a standard arctangent integral:

$$2I = [\arctan(x)]_{1/\sqrt{3}}^{\sqrt{3}} = \arctan(\sqrt{3}) - \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

Dividing by 2, we get the final answer:

$$I = \frac{\pi}{12}$$

1.2 Hard Integral

Rational Arctangent Symmetry

Evaluate the integral:

$$\int_0^1 \frac{\sqrt{3} \tan^{-1}\left(\frac{1-x}{x}\right)}{x^2 - x + 1} dx$$

Solution: Let $I = \int_0^1 \frac{\sqrt{3} \arctan\left(\frac{1-x}{x}\right)}{x^2 - x + 1} dx$. We apply the property of definite integrals $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. Substituting $x \rightarrow 1-x$: The denominator becomes: $(1-x)^2 - (1-x) + 1 = 1 - 2x + x^2 - 1 + x + 1 = x^2 - x + 1$, meaning it is invariant. The argument of the arctangent becomes: $\frac{1-(1-x)}{1-x} = \frac{x}{1-x}$. So the integral can be written as:

$$I = \int_0^1 \frac{\sqrt{3} \arctan\left(\frac{x}{1-x}\right)}{x^2 - x + 1} dx$$

Add this new expression to the original integral:

$$2I = \int_0^1 \frac{\sqrt{3} \left[\arctan\left(\frac{1-x}{x}\right) + \arctan\left(\frac{x}{1-x}\right) \right]}{x^2 - x + 1} dx$$

For any $z > 0$, $\arctan(z) + \arctan(1/z) = \frac{\pi}{2}$. Since $x \in (0, 1)$, $z = \frac{1-x}{x} > 0$. Thus, the numerator simplifies to a constant $\sqrt{3} \frac{\pi}{2}$:

$$2I = \frac{\sqrt{3}\pi}{2} \int_0^1 \frac{1}{x^2 - x + 1} dx$$

Complete the square in the denominator: $x^2 - x + 1 = (x - \frac{1}{2})^2 + \frac{3}{4}$. Let $u = x - \frac{1}{2}$, so $du = dx$, with bounds changing from $[-1/2, 1/2]$:

$$\int_{-1/2}^{1/2} \frac{1}{u^2 + (\sqrt{3}/2)^2} du = \left[\frac{2}{\sqrt{3}} \arctan\left(\frac{2u}{\sqrt{3}}\right) \right]_{-1/2}^{1/2}$$

Evaluate the bounds:

$$\frac{2}{\sqrt{3}} \left(\arctan\left(\frac{1}{\sqrt{3}}\right) - \arctan\left(-\frac{1}{\sqrt{3}}\right) \right) = \frac{2}{\sqrt{3}} \left(\frac{\pi}{6} - \left(-\frac{\pi}{6}\right) \right) = \frac{2}{\sqrt{3}} \left(\frac{\pi}{3} \right) = \frac{2\pi}{3\sqrt{3}}$$

Substitute this back into the equation for $2I$:

$$2I = \frac{\sqrt{3}\pi}{2} \left(\frac{2\pi}{3\sqrt{3}} \right) = \frac{\pi^2}{3}$$

Dividing by 2 yields:

$$I = \frac{\pi^2}{6}$$

Differentials

2.1 Medium Differential

First-Order Linear ODE with Inverse Cotangent

Solve the first-order linear differential equation:

$$\frac{dy}{dx} + \frac{y}{x^2 + 1} = e^{\cot^{-1}(x)}$$

Given the initial condition $y(0) = e^{i\pi}$, where $i = \sqrt{-1}$, find the exact value of $y(1)$.

Solution: We recognize this as a first-order linear ordinary differential equation of the form $\frac{dy}{dx} + P(x)y = Q(x)$. We find the integrating factor $\mu(x)$:

$$\mu(x) = \exp\left(\int P(x) dx\right) = \exp\left(\int \frac{1}{x^2 + 1} dx\right) = e^{\arctan(x)}$$

Multiply the entire differential equation by the integrating factor:

$$e^{\arctan(x)} \frac{dy}{dx} + \frac{e^{\arctan(x)}}{x^2 + 1} y = e^{\cot^{-1}(x)} e^{\arctan(x)}$$

The left side collapses by the product rule into $\frac{d}{dx} [ye^{\arctan(x)}]$. For the right side, recall the identity $\arctan(x) + \cot^{-1}(x) = \frac{\pi}{2}$:

$$e^{\cot^{-1}(x)} e^{\arctan(x)} = e^{\cot^{-1}(x) + \arctan(x)} = e^{\pi/2}$$

The equation is now:

$$\frac{d}{dx} [ye^{\arctan(x)}] = e^{\pi/2}$$

Integrate both sides with respect to x :

$$ye^{\arctan(x)} = xe^{\pi/2} + C$$

Apply the initial condition $y(0) = e^{i\pi}$. By Euler's identity, $e^{i\pi} = -1$, so $y(0) = -1$.

$$(-1)e^{\arctan(0)} = 0 \cdot e^{\pi/2} + C \implies -1 \cdot 1 = C \implies C = -1$$

The particular solution is:

$$ye^{\arctan(x)} = xe^{\pi/2} - 1$$

To find $y(1)$, substitute $x = 1$:

$$y(1)e^{\arctan(1)} = (1)e^{\pi/2} - 1 \implies y(1)e^{\pi/4} = e^{\pi/2} - 1$$

Divide by $e^{\pi/4}$:

$$y(1) = \frac{e^{\pi/2} - 1}{e^{\pi/4}} = e^{\pi/4} - e^{-\pi/4}$$

This can be expressed exactly using the hyperbolic sine function:

$$y(1) = 2 \sinh\left(\frac{\pi}{4}\right)$$

2.2 Hard Differential

Exact Equation via Integrating Factor

Solve the differential equation:

$$\frac{y(1 + 2x^2y)}{x(3 + 4x^2y)} + \frac{1}{2} \frac{dy}{dx} = 0, \quad \text{for } x > 0, y > 0$$

If the curve passes through the point $(1, 1)$, find the value of $y^3(1 + 4y)$ at $x = 2$.

Solution: Rearrange the equation into the standard differential form $M(x, y) dx + N(x, y) dy = 0$:

$$2y(1 + 2x^2y) dx + x(3 + 4x^2y) dy = 0$$

Let $M = 2y + 4x^2y^2$ and $N = 3x + 4x^3y$. Check for exactness:

$$\frac{\partial M}{\partial y} = 2 + 8x^2y, \quad \frac{\partial N}{\partial x} = 3 + 12x^2y$$

Since $M_y \neq N_x$, it is not exact. We search for an integrating factor of the form $\mu(x, y) = x^a y^b$. Multiplying M and N by μ :

$$M' = 2x^a y^{b+1} + 4x^{a+2} y^{b+2}, \quad N' = 3x^{a+1} y^b + 4x^{a+3} y^{b+1}$$

For exactness, we require $M'_y = N'_x$:

$$2(b+1)x^a y^b + 4(b+2)x^{a+2} y^{b+1} = 3(a+1)x^a y^b + 4(a+3)x^{a+2} y^{b+1}$$

Equating coefficients of like terms:

$$2(b+1) = 3(a+1) \implies 2b - 3a = 1$$

$$4(b+2) = 4(a+3) \implies b+2 = a+3 \implies b-a = 1 \implies b = a+1$$

Substitute $b = a+1$ into the first equation:

$$2(a+1) - 3a = 1 \implies 2a + 2 - 3a = 1 \implies -a = -1 \implies a = 1$$

Thus $b = 2$. The integrating factor is $\mu = xy^2$. Multiply the original ODE by xy^2 :

$$(2xy^3 + 4x^3y^4) dx + (3x^2y^2 + 4x^4y^3) dy = 0$$

Now it is exact. There exists a function $F(x, y)$ such that $F_x = M'$ and $F_y = N'$. Integrate M' with respect to x :

$$F(x, y) = \int (2xy^3 + 4x^3y^4) dx = x^2y^3 + x^4y^4 + g(y)$$

Differentiate with respect to y and equate to N' :

$$F_y = 3x^2y^2 + 4x^4y^3 + g'(y) = 3x^2y^2 + 4x^4y^3 \implies g'(y) = 0$$

The general solution is $x^2y^3 + x^4y^4 = C$. Apply the initial condition $x = 1, y = 1$:

$$(1)^2(1)^3 + (1)^4(1)^4 = C \implies 1 + 1 = C \implies C = 2$$

The implicit equation of the curve is $x^2y^3 + x^4y^4 = 2$. To find $y^3(1 + 4y)$ at $x = 2$, substitute $x = 2$ into the curve equation:

$$(2)^2y^3 + (2)^4y^4 = 2 \implies 4y^3 + 16y^4 = 2$$

Divide the entire equation by 4:

$$y^3 + 4y^4 = \frac{2}{4} = \frac{1}{2}$$

Factor out y^3 :

$$y^3(1 + 4y) = 0.5$$

Derivatives

3.1 Medium Derivative

Optimization of a Rectangle's Diagonal

Let $k \in \mathbb{R}$. Consider a rectangle with sides measuring k^{2026} and $\frac{1}{k^{2026}}$. Among all possible values of k , determine the minimum value of the diagonal of this rectangle.

Solution: Let the sides of the rectangle be $a = k^{2026}$ and $b = \frac{1}{k^{2026}}$. (Assume $k \neq 0$ so the rectangle is well-defined). The diagonal D is given by the Pythagorean theorem:

$$D = \sqrt{a^2 + b^2} = \sqrt{(k^{2026})^2 + \left(\frac{1}{k^{2026}}\right)^2} = \sqrt{k^{4052} + k^{-4052}}$$

To find the minimum value of the expression under the radical, we can apply the AM-GM (Arithmetic Mean-Geometric Mean) inequality, which states that for any non-negative real numbers x and y , $x + y \geq 2\sqrt{xy}$. Let $x = k^{4052}$ and $y = k^{-4052}$:

$$k^{4052} + \frac{1}{k^{4052}} \geq 2\sqrt{k^{4052} \cdot \frac{1}{k^{4052}}} = 2\sqrt{1} = 2$$

Equality holds when $k^{4052} = 1$, which is true for $k = 1$ or $k = -1$. Therefore, the minimum value of D^2 is 2, meaning the minimum diagonal length is:

$$D_{\min} = \sqrt{2}$$

3.2 Hard Derivative

Roots of a Shifted Taylor Polynomial

Consider a third-degree polynomial $f(x)$. Given that:

$$f(2) = 1, \quad f'(2) = 1, \quad f''(2) = 2, \quad f'''(2) = 6.$$

Let a, b, c, d satisfy the equation $(x - 1)f(x + 2) = 0$. Find the value of $(a + 2)(b + 2)(c + 2)(d + 2)$.

Solution: Because $f(x)$ is a third-degree polynomial, it can be exactly represented by its finite Taylor series expansion centered at $x = 2$:

$$f(x) = f(2) + f'(2)(x - 2) + \frac{f''(2)}{2!}(x - 2)^2 + \frac{f'''(2)}{3!}(x - 2)^3$$

Substituting the given derivative values:

$$f(x) = 1 + 1(x - 2) + \frac{2}{2}(x - 2)^2 + \frac{6}{6}(x - 2)^3 = 1 + (x - 2) + (x - 2)^2 + (x - 2)^3$$

The equation is $(x - 1)f(x + 2) = 0$. Let's evaluate $f(x + 2)$ by substituting $x + 2$ into our polynomial:

$$f(x + 2) = 1 + ((x + 2) - 2) + ((x + 2) - 2)^2 + ((x + 2) - 2)^3 = 1 + x + x^2 + x^3$$

Thus, the equation for the roots a, b, c, d becomes:

$$(x - 1)(1 + x + x^2 + x^3) = 0$$

Recognizing the factorization of a difference of powers, this simplifies perfectly to:

$$x^4 - 1 = 0$$

The roots of this polynomial are the fourth roots of unity: $1, -1, i, -i$. We are tasked with evaluating the product $P = (a + 2)(b + 2)(c + 2)(d + 2)$. Consider the polynomial $P(x) = x^4 - 1 = (x - a)(x - b)(x - c)(x - d)$. If we substitute $x = -2$:

$$P(-2) = (-2 - a)(-2 - b)(-2 - c)(-2 - d) = (-1)^4(2 + a)(2 + b)(2 + c)(2 + d)$$

$$P(-2) = (a + 2)(b + 2)(c + 2)(d + 2)$$

Evaluate $P(-2)$ directly from the polynomial equation:

$$P(-2) = (-2)^4 - 1 = 16 - 1 = 15$$

Thus, $(a + 2)(b + 2)(c + 2)(d + 2) = 15$.

Physics & Engineering

4.1 Mechanics

Monatomic Gas Kinetic Energy

A monatomic ideal gas is confined to a container at a temperature of $T = 300$ K. What is the average translational kinetic energy of an atom of the gas? Give your answer in terms of 10^{-21} J. Round to two decimal places. Take $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$.

Solution: According to the kinetic theory of gases, the average translational kinetic energy E_k of a single molecule/atom in a gas is determined by its absolute temperature T using the equipartition theorem (with 3 degrees of freedom for a monatomic gas):

$$E_k = \frac{3}{2} k_B T$$

Substitute the given Boltzmann constant k_B and temperature $T = 300$ K:

$$E_k = 1.5 \times (1.38 \times 10^{-23} \text{ J K}^{-1}) \times 300 \text{ K}$$

$$E_k = 450 \times 1.38 \times 10^{-23} \text{ J} = 621 \times 10^{-23} \text{ J}$$

Convert the scientific notation to match the requested format of 10^{-21} J:

$$E_k = 6.21 \times 10^{-21} \text{ J}$$

The value is 6.21.

4.2 Quantum Mechanics

Superposition in a 1D Box

A particle of mass m is confined to a one-dimensional square well of width L . The potential $V(x)$ is 0 if $0 < x < L$ and ∞ otherwise. The particle is in a normalized state:

$$\psi(x) = \sqrt{\frac{8}{5L}} \left[\sin\left(\frac{\pi x}{L}\right) + \frac{1}{2} \sin\left(\frac{3\pi x}{L}\right) \right]$$

Find the absolute difference between the expectation value of the energy and the energy of the first excited state, expressed as a multiple k of the ground-state energy E_1 . Calculate the exact numerical value of k .

Solution: The stationary states of the infinite square well are $\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$ with eigenenergies $E_n = n^2 E_1$, where $E_1 = \frac{\pi^2 \hbar^2}{2mL^2}$. We rewrite the given state $\psi(x)$ as a linear combination of these basis states:

$$\psi(x) = c_1 \phi_1(x) + c_3 \phi_3(x)$$

To find c_1 , we divide the coefficient of $\sin(\frac{\pi x}{L})$ by $\sqrt{\frac{2}{L}}$:

$$c_1 = \frac{\sqrt{8/(5L)}}{\sqrt{2/L}} = \sqrt{\frac{8}{10}} = \sqrt{\frac{4}{5}} \implies |c_1|^2 = \frac{4}{5}$$

Similarly, for c_3 :

$$c_3 = \frac{\frac{1}{2}\sqrt{8/(5L)}}{\sqrt{2/L}} = \frac{1}{2}\sqrt{\frac{4}{5}} = \frac{1}{\sqrt{5}} \implies |c_3|^2 = \frac{1}{5}$$

The expectation value of the energy is the sum of the probabilities multiplied by their respective energy eigenvalues:

$$\langle E \rangle = |c_1|^2 E_1 + |c_3|^2 E_3 = \frac{4}{5} E_1 + \frac{1}{5} (3^2 E_1) = \frac{4}{5} E_1 + \frac{9}{5} E_1 = \frac{13}{5} E_1 = 2.6 E_1$$

The first excited state corresponds to $n = 2$, with energy E_2 :

$$E_2 = 2^2 E_1 = 4 E_1$$

The absolute difference between $\langle E \rangle$ and E_2 is:

$$|\langle E \rangle - E_2| = |2.6 E_1 - 4 E_1| = |-1.4 E_1| = 1.4 E_1$$

The exact numerical value of k is 1.4 (or $\frac{7}{5}$).

4.3 Electromagnetism

Resistor Power Dissipation

Three identical resistors X , Y , and Z are connected to a battery. Resistors Y and Z are in parallel, and resistor X is in series with the parallel combination of Y and Z . The battery supplies a total power of $P = 24$ W. Find the power dissipated in resistor Z (in W). Assume wires and battery have negligible resistance.

Solution: Let the resistance of each identical resistor be R . Since Y and Z are in parallel, their equivalent resistance R_{YZ} is:

$$R_{YZ} = \left(\frac{1}{R} + \frac{1}{R} \right)^{-1} = \frac{R}{2}$$

This parallel combination is in series with X , so the total equivalent resistance R_{eq} of the circuit is:

$$R_{eq} = R_X + R_{YZ} = R + \frac{R}{2} = \frac{3R}{2}$$

The total power P supplied by the battery is $I^2 R_{eq} = 24$ W, where I is the total current leaving the battery. The current I passes entirely through resistor X . Thus, the power dissipated by X is:

$$P_X = I^2 R$$

We can relate P_X to the total power:

$$P_{\text{total}} = I^2 \left(\frac{3R}{2} \right) = 1.5(I^2 R) = 24 \implies I^2 R = \frac{24}{1.5} = 16 \text{ W}$$

So, X dissipates 16 W. By conservation of energy, the remaining power is dissipated by the parallel combination Y and Z:

$$P_{YZ} = P_{\text{total}} - P_X = 24 - 16 = 8 \text{ W}$$

Since resistors Y and Z are identical and in parallel, they share the voltage equally and have the same resistance, meaning the current I splits perfectly in half ($\frac{I}{2}$ through each). Consequently, they dissipate equal amounts of power.

$$P_Z = \frac{P_{YZ}}{2} = \frac{8}{2} = 4 \text{ W}$$